# FINSHIELD

## Loan Default Risk Analysis & Borrower Segmentation

DVA Capstone 2 | Team DVA-G1 | Section D
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github.com/dakshbatra01/SectionD_G1_FinShield
Tableau Public Dashboard

# Single-Metric Credit Screening Fails — ₹1.12B in Defaults Signals Systemic Risk

## SECTOR CONTEXT

Retail lending faces high default risk (NPAs). Traditional screening relies heavily on credit scores, which are not tailored to specific portfolios. Result: Weak risk prediction and high financial exposure.

## PROBLEM STATEMENT

How can we identify high-risk borrowers before approval using data-driven indicators instead of relying on a single credit score?

## OBJECTIVE

To identify high-risk borrowers before loan approval using data-driven indicators (LTV, DTI, and behavioral factors) in order to reduce default rates and financial losses.

## PROJECT SCOPE

- 15,000 loan records (2019)
- Python ETL + EDA + statistical analysis
- 3 Tableau dashboards
- 5 business recommendations

**₹1.12B**

Exposure at Default (22.64% of ₹4.97B portfolio)

**23.72%**

Portfolio Default Rate (3,558 of 15,000 loans)

**37.79%**

High Risk Borrowers (High + Very High tiers)

# 15,000 Loan Records Across 5 Python Notebooks Power a Reproducible ETL Pipeline

## DATASET OVERVIEW

**Source:** Kaggle Loan Default Dataset

**Rows:** 15,000 (one per loan)

**Raw Columns:** 34

**Tableau-Ready:** 37 (3 derived)

**Period:** 2019 snapshot

**Target:** status (0=No Default, 1=Default)

**Default Rate:** 23.72%

## KEY DATA CLEANING (PYTHON ETL)

- Converted key variables (status, income, loan_amount) into numeric formats to enable accurate aggregation and KPI computation

- Handled missing values in critical fields (income, DTI) using appropriate imputation techniques to ensure data completeness and avoid bias in risk analysis

- Standardized and renamed categorical variables (e.g., type1/type2 → loan_type, dtir1 → debt_to_income_ratio) to improve consistency and interpretability across analysis and dashboards

## ETL PIPELINE — 5 JUPYTER NOTEBOOKS

01 Extraction & Cleaning — 02 Transformation — 03 EDA — 04 Statistical Analysis — 05 Load

## KEY VARIABLES

| Column | Type | Description | Role |
|---|---|---|---|
| status | int | Target: 0=No Default, 1=Default | Target variable |
| ltv | float | Loan-to-Value ratio | KPI / Risk Score |
| debt_to_income | float | DTI ratio (renamed from dtir1) | KPI / Risk Score |
| credit_score | int | Bureau score (p=0.735) | Statistical testing |
| risk_tier | string | Derived: Low/Med/High/VH | Lending Decision |

# Portfolio Default Rate of 23.72% Demands a Structured KPI Framework at Every Level

| 23.72% | ₹4.97B | 37.79% | ₹1.12B | 4.036% |
|---|---|---|---|---|
| Portfolio Default Rate | Total Loan Portfolio | High Risk Borrowers % | Exposure at Default | Average Interest Rate |

## KPI FRAMEWORK — ALL METRICS

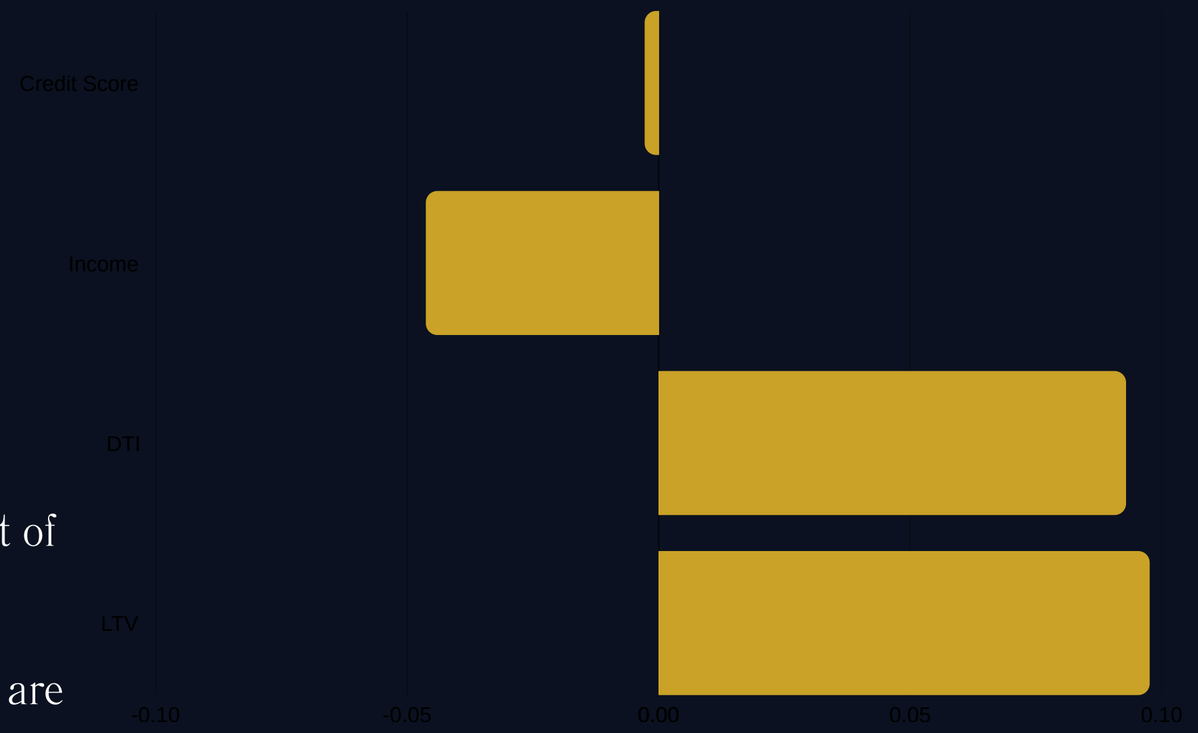
| KPI | Formula | Why These KPIs | How they connect |
|---|---|---|---|
| Total Loans | COUNT(*) | Defines portfolio size and scale | Defines portfolio size and scale |
| Default Rate | Defaults / Total × 100 | Core indicator of portfolio risk | Drives Expected Loss and risk segmentation |
| Total Loan Amount | SUM(loan_amount) | Measures total financial exposure | Combined with default rate → shows loss magnitude |
| High Risk % | COUNT(LTV>80) / Total × 100 | Identifies risky borrower segment | Explains why default rate is high |
| Expected Loss (EL) | Default Rate × LGD × EAD | Quantifies total financial risk | Combines default + exposure → key decision metric |
| High Risk Borrowers % | COUNT(LTV>80) / Total × 100 | Identifies proportion of high-risk borrowers | Explains why default rate is high and feeds into Expected Loss |
| Borrower Income Gap | AVG(income, defaulters) / AVG(income, all) | Shows repayment capacity difference | Explains borrower-level default behavior |
| Most Risky Loan Type | MAX(Default Rate BY loan_type) | Identifies high-risk products | Links product strategy to default rate |
| Highest Loss Region | MAX(SUM(defaulted_amount) BY region) | Detects geographic risk concentration | Connects location → loss exposure |
| Top Loan Purpose | MAX(SUM(defaulted_amount) BY purpose) | Finds major loss driver segment | Explains where most defaults originate |

# LTV and DTI Show Clear Default Separation — Credit Score Shows Near-Zero Correlation

## KEY EDA FINDINGS

1. **23.72% default rate** — nearly 1 in 4 loans. ₹1.12B concentrated in defaulted accounts demands structural intervention.
2. **LTV gap:** Defaulters carry mean LTV 75.78 vs 71.65 for performing (4.1-point gap). **DTI gap:** 39.59 vs 37.49 (2.1-point gap).
3. **North-East region** default rate ~52% above North — highest of all four segments, indicating poor underwriting quality.
4. **Correlation heatmap:** LTV (r=0.0976) and DTI (r=0.0929) are highest predictors. Credit score r=**-0.0028** — effectively zero.
5. **Personal loans default at 33.49%** — approximately 54% higher than home loans at 21.74%.

## CORRELATION WITH DEFAULT STATUS




Note: Credit score correlation of -0.0028 is effectively zero — confirming its limited predictive value in this sample. All values from notebooks/03_eda.ipynb and 04_statistical_analysis.ipynb.

# Statistical Testing Validates LTV as the Strongest Predictor — Each SD Increase Raises Default Odds by 25.1%

## LOGISTIC REGRESSION — ODDS RATIOS

Odds ratio > 1 indicates higher default likelihood. LTV odds ratio of 1.251 means each 1 SD increase is associated with a 25.1% increase in default odds — the highest observed effect among continuous predictors.

## CHI-SQUARED TESTS — CATEGORICAL

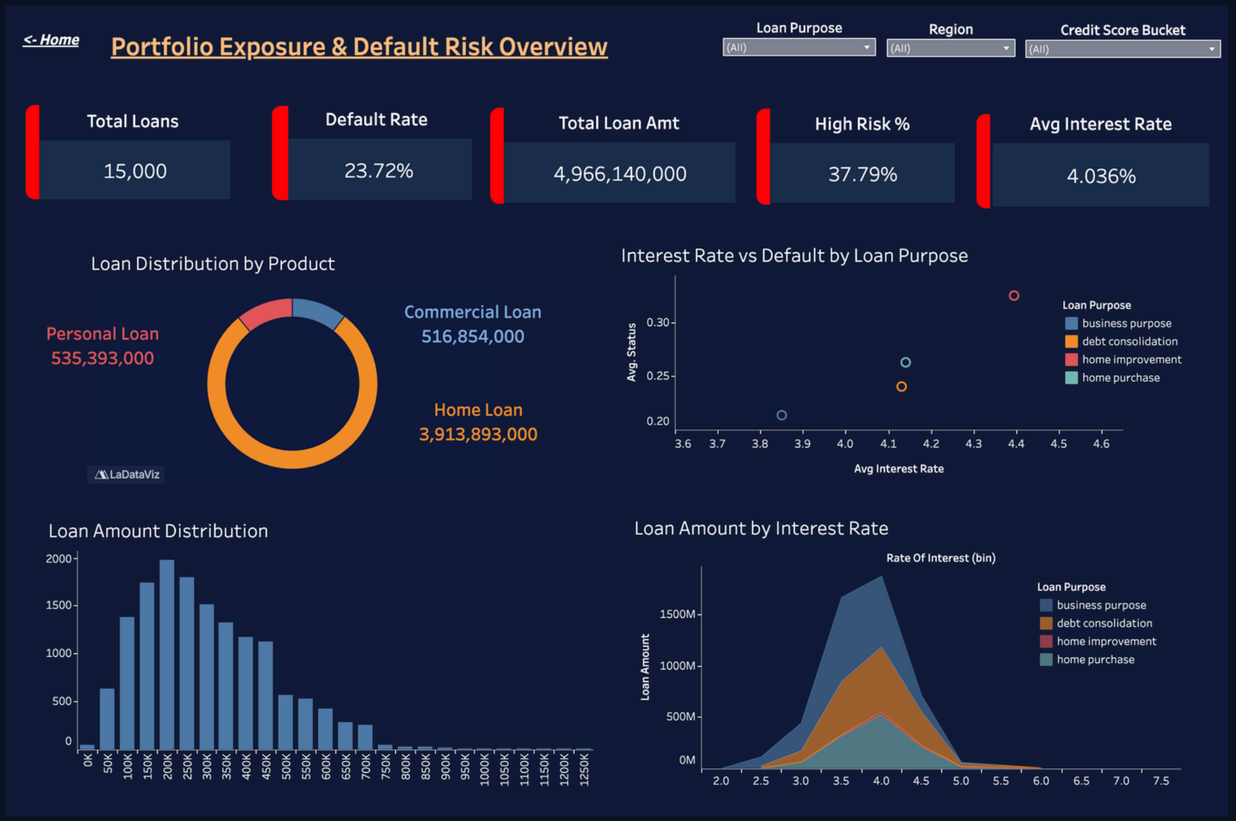
| Feature | $\chi^2$ Value | p-value | Verdict |
|---|---|---|---|
| lump_sum_payment | 498.5 | < 0.001 | Significant |
| ltv_risk_bucket | 491.4 | < 0.001 | Significant |
| neg_ammortization | 363.1 | < 0.001 | Significant |
| loan_type | — | < 0.05 | Significant |
| region | — | < 0.05 | Significant |
| credit_score_bucket | 2.13 | 0.722 | Not Sig. |

Product-level features demonstrate materially stronger association with default than credit score bands — supporting a reorientation toward structural loan and borrower financial characteristics. $\alpha = 0.05$ throughout.

# Three-Dashboard Tableau Suite Surfaces Portfolio, Borrower, and Geographic Risk Intelligence

Tableau Public: [public.tableau.com/app/profile/daksh.batra/viz/shared/4SBGTH5PQ](public.tableau.com/app/profile/daksh.batra/viz/shared/4SBGTH5PQ)

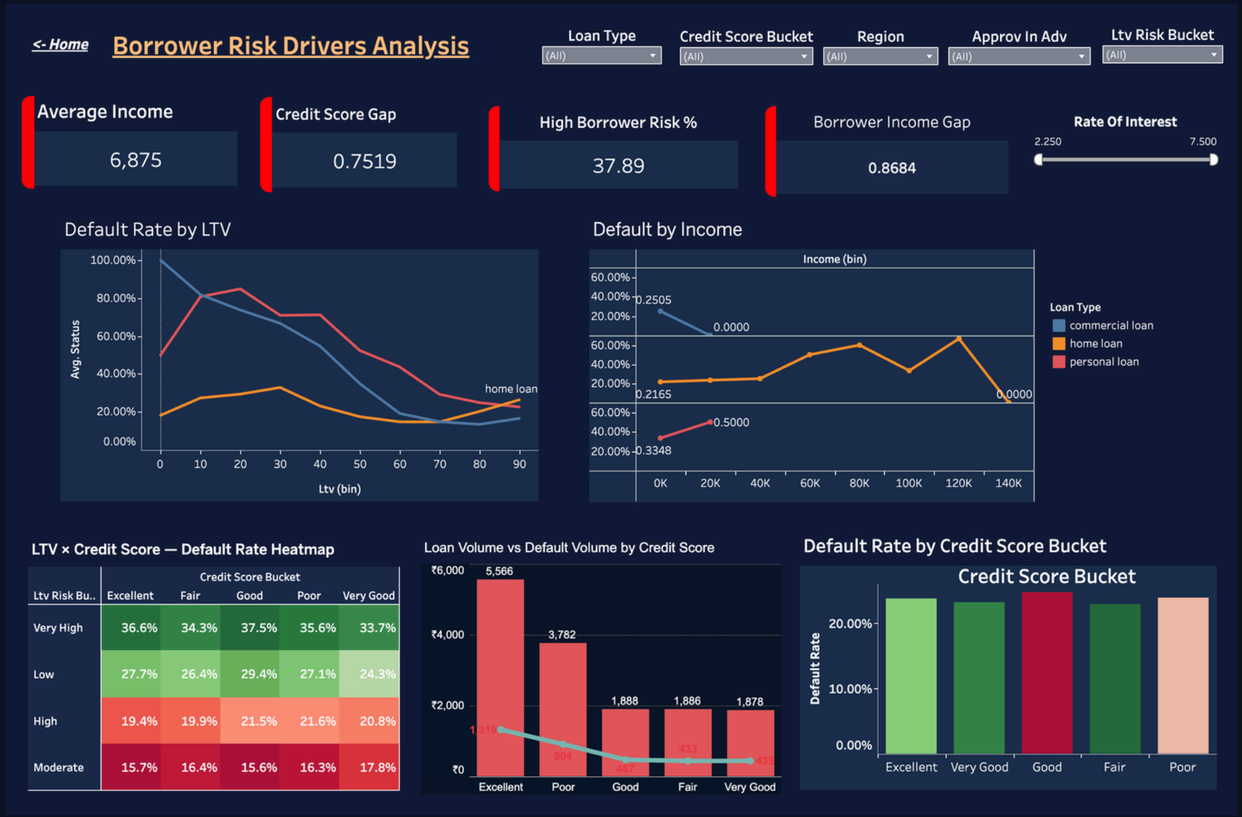
**Navigation Home** — Entry point with three dashboard cards



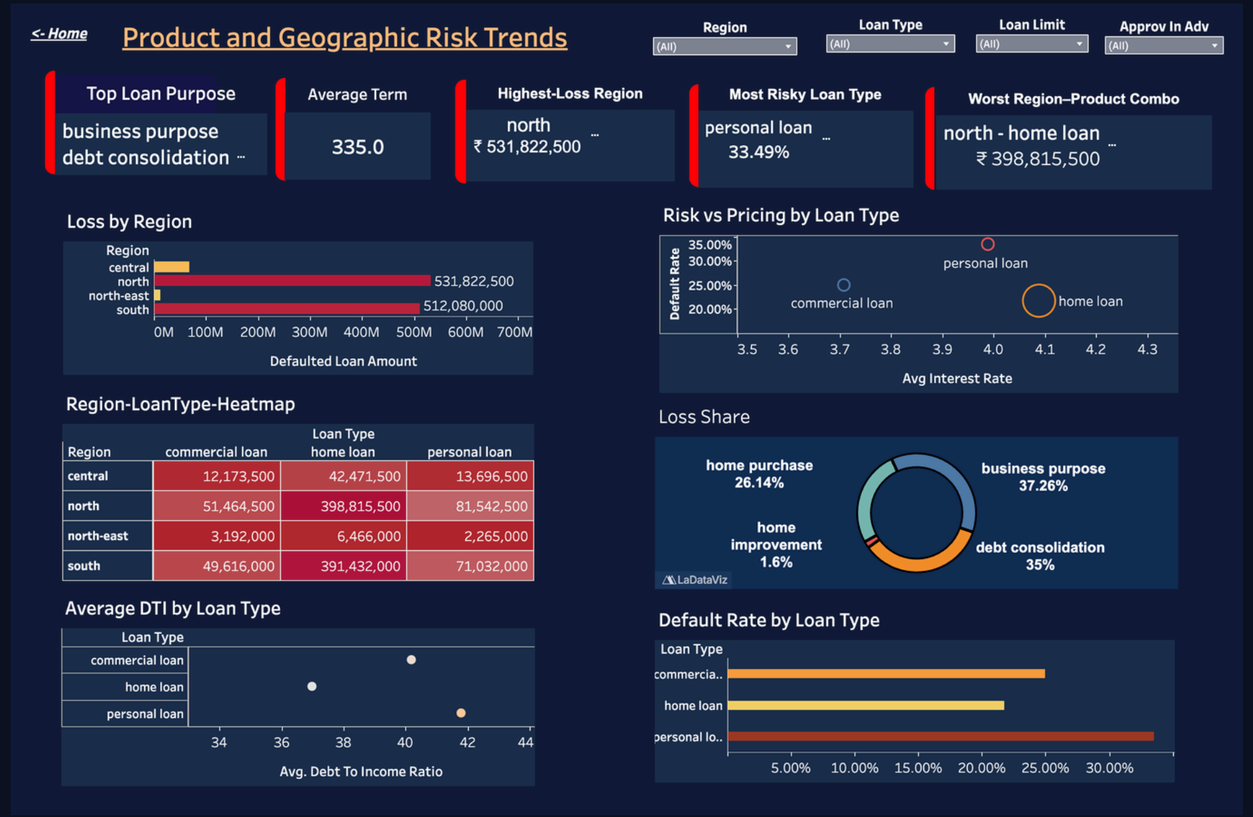

**Dashboard 1** — Portfolio Exposure & Default Risk Overview (CRO View)

# Three-Dashboard Tableau Suite Surfaces Portfolio, Borrower, and Geographic Risk Intelligence

Tableau Public: [public.tableau.com/app/profile/daksh.batra/viz/shared/4SBGTH5PQ](public.tableau.com/app/profile/daksh.batra/viz/shared/4SBGTH5PQ)




Dashboard 2 — Borrower Risk Drivers Analysis (Lending Team View)




Dashboard 3 — Product & Geographic Risk Trends (Strategy View)

# Five Prioritised Recommendations — From Statistical Evidence to Actionable Risk Policy

| Priority | Recommendation | Expected Impact | Difficulty | Evidence |
|---|---|---|---|---|
| HIGH | Implement dual LTV + DTI approval policy to replace single-metric credit screening | Addresses the two leading predictors. Dashboard 2 confirms credit score adds no differentiation within same LTV tier. | Medium | LTV OR=1.251, DTI OR=1.216 |
| HIGH | Restrict or eliminate negative amortisation structures for Medium/High risk tiers | Targets the strongest categorical risk signal ($\chi^2$=363.1, p<0.001). | Medium | $\chi^2$=363.1 |
| HIGH | Reprice personal loans upward — they are underpriced relative to their 33.49% default rate | Improves risk-adjusted return immediately. Dashboard 3 Risk vs Pricing scatter confirms underpricing. | Low | 33.49% vs 21.74% |
| MEDIUM | Establish geographic monitoring for North (₹531.8M loss) and North-East (33.60% default rate) | Mitigates geographic concentration risk. Highest absolute loss + highest default rate regions. | Low | Dashboard 3 |
| LOW | Build ML-based borrower scoring model using validated features (LTV, DTI, neg_amm, income, lump_sum) | High (longer term) — enables individual probability scoring at origination. | High | All validated features |

# Combined Policy Implementation Could Reduce Portfolio Defaults by 25–40%, Saving ₹280M–₹448M

## DIRECTIONAL IMPACT SCENARIOS

| Intervention | Default Reduction | Scenario Savings |
|---|---|---|
| LTV + DTI dual-band screening | 15–25% | ₹168M – ₹280M |
| Restrict negative amortisation | 8–12% | ₹90M – ₹134M |
| Personal loan repricing | Revenue improvement | Risk-adjusted margin |
| Geographic exposure controls | 5–8% | ₹56M – ₹90M |
| Combined policy | **25–40%** | **₹280M – ₹448M** |

All savings figures are directional scenario estimates derived from statistical effect sizes. Treat as conservative order-of-magnitude indicators, not guaranteed results. Cost of inaction: ₹1.12B already at risk, 37.79% of portfolio classified as high risk.

## ESTIMATED SAVINGS BY INTERVENTION

Mid-point estimates shown. Conservative assumptions applied. Personal loan repricing excluded as it improves revenue rather than reducing defaults.

# Single-Year Cross-Sectional Data Limits Generalisation — ML Scoring and Real-Time Integration Are the Next Frontier

## LIMITATIONS

**1. Data & Time Scope Constraints**

Dataset limited to a single origination year (2019)– no trend or cycle comparison

Absence of macroeconomic variables (unemployment, rate cycle, housing index)

**2. Data Quality Considerations**

Median imputation applied to missing income, property value, and interest fields

Large "Unknown" gender segment limits demographic precision

**4. Association, not causation:** All findings indicate statistical association within this sample. Results support decisions, not causal proofs.

**5. Class imbalance:** The 23.72% default rate creates moderate imbalance, constraining logistic regression AUC. Production modelling would require SMOTE or cost-sensitive learning.

## FUTURE SCOPE & NEXT STEPS

**1. ML Scoring Model:** Random Forest or XGBoost using validated features (LTV, DTI, neg_ammortization, lump_sum, income) — enabling borrower-level default probability scoring at origination.

**2. Alternative Data Integration:** Bank statements, GST filings, utility payment history, UPI transaction patterns — aligned with India's OCEN framework.

**3. Multi-Year Temporal Analysis:** Extending to 2016–2023 would allow testing predictor relationships across economic cycles.

**4. Real-Time API Integration:** Deploy validated risk scoring as a REST API endpoint for loan origination system integration.

**5. Live Tableau Refresh:** Connect to live or incremental data source for near-real-time monitoring of default rates and risk tier distributions.

**Ethics Note:** Analytical models should support — not replace — human lending decisions. Risk frameworks must not contribute to discriminatory practices. Demographic variables such as gender should be handled with regulatory caution. All scoring tools subject to regular fairness audits.

# Team DVA-G1 — Seven Members, Seven Specialisations, One Reproducible Pipeline

## Daksh Batra
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Dataset, ETL, EDA, Statistical Analysis, Tableau

## Prakhar Rawat
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Dataset sourcing, data dictionary, quality assurance

## Nitin Kumar
**ETL Lead** | @Nitin-0017

Extraction, cleaning, transformation pipeline

## Isha Tomar
**Analysis Lead** | @Bytebard089

EDA, statistical analysis, insight generation

## Kapish Rohilla
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Tableau dashboard suite, visual design

## Rishi Raj
**Strategy Lead** | @rishiraj38

Recommendations, impact estimation, strategy

## Nishant Ranjan Singh
**PPT & Quality Lead** | @IAmNishantSingh

Presentation design, report, quality assurance

**Repository Metrics:** 45+ commits | 6 contributors | 5 Jupyter notebooks | 3-dashboard Tableau suite | Full data dictionary | All contributions verifiable through GitHub Insights and PR history

# THANK YOU

**FinShield — Converting Raw Data Into Practical Risk Intelligence**

github.com/dakshbatra01/SectionD_G1_FinShield

Tableau Public Dashboard

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